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2026 Annual Symposium

#AMIA2026



Causal AI for Health: Moving from Prediction to Actionable Inference

W18: AMIA 2026 KDDM Working Group Collaborative Workshop

Zhe He, PhD, FACMI, FIAHSI, FAMIA

Professor, School of Information, Florida State University
Director, Institute for Successful Longevity, Florida State University
Informatics Lead, UF-FSU Clinical and Translational Science Award Hub
Chair, AMIA Knowledge Discovery and Data Mining (KDDM) Working Group

Saturday, November 7, 2026 · 1:00–4:30 PM · Dallas, TX

Website

Workshop Organizers

Zhe He, PhD, FACMI, FIAHSI, FAMIA
Florida State University
Chair, AMIA KDDM WG · Workshop lead and panel moderator

Hilda B. Klasky, FAMIA
Oak Ridge National Laboratory
Vice Chair, AMIA KDDM WG · Workshop logistics

Mattia Prosperi, PhD, FAMIA, FACMI
University of Florida
Associate Dean for AI and Innovation · Panelist

Carl Yang, PhD
Emory University
Assistant Professor of Computer Science

Humayera Islam, PhD
University of Chicago
Secretary, AMIA KDDM WG · Poster session coordinator

Navya Martin Kollapally, PhD
Kean University
Poster session

Jakir Hossain Bhuiyan Masud, PhD
University of Alabama at Birmingham
Workshop logistics

Thank you
to our speakers, panelists, and poster presenters

Disclosure of Relevant Financial Relationships

The following presenter(s) **have no** relevant financial relationship(s) with ineligible companies to disclose.

- Zhe He, PhD
- Yi Guo, PhD
- Nansu Zong, PhD
- Vincent Jeanselme, PhD
- Mattia Prosperi, PhD
- C. Mahony Reategui-Rivera, PhD
- Yves Lussier, PhD
- Carolyn Scheese, PhD

*All of the relevant financial relationships listed for these individuals have been mitigated.

Learning Objectives

At the conclusion of this presentation the learner will be able to:

1. Distinguish causal inference from predictive modeling and determine when causal methods are required for clinical or policy decisions
2. Identify key sources of bias and confounding in observational health data and select appropriate study designs, such as target trial emulation, for estimating treatment effects
3. Describe emerging approaches that integrate structural causal models and generative AI with machine learning to improve fairness, transportability, and real-world deployment

Welcome from the AMIA KDDM Working Group

A collaborative workshop bringing together informaticians, clinicians, epidemiologists, biostatisticians, and AI scientists to advance trustworthy, actionable AI in health.

Who is in the room (expected mix)

25% — **Novice**
Foundations: DAGs, counterfactual reasoning, confounding

50% — **Intermediate**
Adding target trial emulation and causal ML to predictive work

25% — **Advanced**
Developing causal AI methods and leading translational or regulatory programs

Building on past KDDM workshops

2025 — **Open-Source LLMs for Health**
AMIA 2025 Annual Symposium · 100+

2024 — **Knowledge-Augmented LLMs**
AMIA 2024 Annual Symposium · 180+

2024 — **Enriching EHRs with SDoH**
AMIA 2024 Informatics Summit · 100+

2021 — **Bridging Data and Knowledge**
AMIA 2021 Annual Symposium · 100+

Why Causal AI? Prediction Alone Is Not Enough

AI models predict outcomes remarkably well from EHRs, multimodal and observational data, but clinical and public health decisions require answers to three different questions:

WHY

Why do outcomes occur?

Mechanisms, etiology, and confounding

WHAT IF

What would happen under alternative interventions?

Counterfactual reasoning

FOR WHOM

Which patients benefit most from specific treatments?

Heterogeneous treatment effects

Risk prediction → **Causal AI: SCMs · target trial emulation · counterfactuals · ML** → **Interventions, guidelines & policy**

What We Will Explore Today

01

Foundations of causal inference for health data

02

Integrating causal modeling with machine learning and deep learning

03

Causal discovery from EHR and multimodal data

04

Generative AI for healthcare decision making

05

Bias, transportability, and fairness in AI systems

06

Real-world implementation and regulatory considerations

07

Case studies of causal AI for precision health

GOAL

Trustworthy AI that informs interventions and policy

Agenda · Saturday, November 7, 2026

Session 1 · 1:00–2:30 PM

1:00 — **Welcome Remarks and Introduction**
Zhe He, Florida State University

1:10 — **Keynote: Causal AI and the Shift from Prediction to Action**
Yi Guo, University of Florida

1:50 — **Target Trial Emulation & Real-World Data**
Nansu Zong, Mayo Clinic

2:10 — **Causal ML & Heterogeneous Treatment Effects**
Vincent Jeanselme, PathomIQ

Coffee Break and Poster: 2:30 PM – 3:00 PM

Session 2 · 3:00–4:30 PM

3:00 — **Hands-on Tutorial on MedGenAI**
Yves Lussier, Carolyn Scheese, C. Mahony Reategui-Rivera, University of Utah

3:45 — **Panel: Can AI Be Trusted Without Causality?**
Moderator: Zhe He · Panelists: Mattia Prosperi, Nansu Zong, Vincent Jeanselme

4:25 — **Closing Remarks & Future Directions**

2:30–3:00 PM Coffee Break & Poster Presentations

Slides

Keynote Speaker

“Causal AI and the Shift from Prediction to Action”



Yi Guo, PhD, FAMIA

1:10 PM · 40 min

Professor, Associate Chair for Data Science, and Chief, Division of Biomedical Informatics and Data Science, University of Florida

- Develops causal modeling and target trial emulation methods to estimate treatment effects from real-world clinical data
- Chief Data Scientist, NCI-designated UF Health Cancer Institute
- Co-Director, Biomedical Informatics Program, UF Clinical and Translational Science Institute

Invited Presentations

1:50 PM · 20 min

Target Trial Emulation & Real-World Data

Nansu Zong, PhD

Associate Professor of Biomedical Informatics, Mayo Clinic

- Builds AI architectures that harmonize real-world evidence with mechanistic biology
- Integrates EHR, knowledge graphs, and pharmacogenomics for therapeutic discovery and outcome prediction
- Chair, AMIA Biomedical Knowledge Representation and Semantics WG

2:10 PM · 20 min

AI for Causal inference, Causal inference for AI

Vincent Jeanselme, PhD

Director of Machine Learning Research, PathomIQ

- Builds and evaluates AI tools for precision medicine, with a focus on their impact on clinical practice and patient outcome
- Works at the intersection of algorithmic safety and clinical decision-making

Slides

Short Presentation

Up next: Short Talk by Nansu Zong, PhD

Causal AI and the Shift from Prediction to Action

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Slides

Short Presentation

Up next: Short Talk by Vincent Jeanselme, PhD

AI for Causal inference

Causal inference for AI

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Poster Session

- Mapping the Social, Behavioral, and Psychosocial Exposome and Incident Mental and Metabolic Health Using Causal Discovery
- Leverage Causal Inference to Quantify the Survival Benefit of Multidisciplinary Care for ALS Using Real-World Data
- Investigating Environmental Exposome Risk Factors of ALS using Double Machine Learning
- Shift the Effectiveness-Equity Frontier Outward: A Causal AI Framework to Prioritize Synergistic Interventions in Improving County-Level Lung Cancer Late-Stage Diagnosis Outcomes
- Evaluation of Causal Reasoning for Large Language Models in Contextualized Clinical Scenarios of Laboratory Test Interpretation

Slides

Hands-on Tutorial on MedGenAI

From text generation to actionable inference

1. **Choose the use case**
   Quick poll: etiologic modeling, differential diagnosis, or interventional decision support

2. **Design and ground**
   Clinical-grade prompt design and retrieval-augmented generation (RAG)

3. **Build an agent**
   Construct GPT-based agents in ChatGPT to prototype causal decision-support and diagnostic tools

4. **Integrate (time permitting)**
   RAG pipelines and EHR integration with n8n; clinician-facing interfaces with Figma AI

PRESENTERS · UNIVERSITY OF UTAH

Yves A. Lussier, MD
Professor and Chair, Biomedical Informatics

Carolyn Scheese, DNP, RN
Associate Professor (Clinical), College of Nursing and DBMI

Christian Mahony Reategui Rivera, MD, MS
PhD candidate, Biomedical Informatics

Bring your laptop
Participants using ChatGPT can build agents along with us

Tutorial



Yves Lussier, PhD, FACMI



Carolyn Scheese, PhD



Christian Mahony Reategui Rivera, PhD

Panel Discussion

“Can AI Be Trusted Without Causality?”

Bias | Fairness | Transportability | Regulatory considerations

MODERATOR
Zhe He, PhD
Florida State University

PANELIST
Mattia Prosperi, PhD
University of Florida

PANELIST
Nansu Zong, PhD
Mayo Clinic

PANELIST
Vincent Jeanselme
PathomIQ

3:45 PM · 40 min · Questions from the audience are welcome

Panel Discussion: Can AI Be Trusted Without Causality?



Zhe He, PhD



Mattia Prosperi, PhD



Vincent Jeanselme, PhD



Nansu Zong, PhD

Panel Discussion: Can AI Be Trusted Without Causality?

Prediction vs. causation — where’s the line?

Panel Discussion: Can AI Be Trusted Without Causality?

From real-world data to clinical insight

Panel Discussion: Can AI Be Trusted Without Causality?

Bias, fairness, and transportability

Panel Discussion: Can AI Be Trusted Without Causality?

Generative AI, regulation, and real-world deployment

Closing Remarks



Zhe He, PhD

Material & Slides